

Step 1: Differential Abundance and Proteome Ranking

To extract stable biological signals from noisy mass spectrometry abundance measurements, the quantified proteome is transformed into a single continuous vector. This step condenses both the magnitude of the fold-change and the statistical confidence of the differential test into a unified metric, sorting the proteins from the most robustly upregulated to the most robustly downregulated.

$$S_i = \text{sign}(\log_2 \text{FC}_i) \times -\log_{10}(P_i)$$

Here, S_i represents the composite ranking metric for protein i . The term $\log_2 \text{FC}_i$ is the empirical Bayes moderated log-fold change of the protein between two experimental conditions, determining the direction (sign) of the biological shift. P_i is the associated p-value derived from the linear model. Taking the negative base-10 logarithm scales high-confidence statistical observations into large magnitudes, ensuring that the most significant driver proteins occupy the extreme ends of the ranked vector.

Step 2: Base Gene Set Enrichment Analysis (GSEA)

Rather than relying on arbitrary significance cutoffs, the pathway enrichment topology is evaluated by executing a weighted random walk down the continuous ranked proteome. This identifies the point of maximum deviation where pathway-specific proteins are most concentrated, defining both the statistical enrichment peak and the core subset of proteins actively driving the biological shift.

$$RES(i) = \sum_{j \leq i, p_j \in K} \frac{|S_j|^p}{N_R} - \sum_{j \leq i, p_j \notin K} \frac{1}{N - |K|}$$

$RES(i)$ is the Running Enrichment Score at rank index i . The variable p_j represents the protein at rank position j , K is the targeted biological pathway, and $|K|$ is the total number of proteins within that pathway. N is the total number of quantified proteins in the background measured universe. N_R is a normalization factor equal to the sum of the absolute ranking metrics $|S_j|^p$ for all proteins in K , and p is a weighting exponent (typically 1). The maximum absolute deviation from zero, $\max_i |RES(i)|$, defines the Enrichment Score (ES). The rank index i^* where this peak occurs isolates the leading-edge core of proteins, denoted as $\mathcal{LE}(K)$.

Step 3: Parametric Noise Injection and Leading-Edge Stability Index

To combat the high variance inherent in small-sample liquid chromatography-mass spectrometry datasets, parametric Gaussian noise is injected directly into the effect sizes. By iteratively recalculating the enrichment walk across these perturbed states, we quantify the absolute invariance of the leading-edge proteins, effectively separating true structural drivers from stochastic boundary artifacts.

$$S_{lesi}(p) = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(p \in \mathcal{LE}_b(K)) \quad \text{where} \quad \epsilon_{i,b} \sim \mathcal{N}(0, \eta \cdot \text{SE}_i^2)$$

$S_{lesi}(p)$ represents the Leading-Edge Stability Index for protein p , bounded between 0 and 1. B denotes the total number of noise-injection iterations, and $\mathbb{I}$ is the indicator function, returning 1 if

protein p falls within the perturbed leading-edge core $\mathcal{LE}_b(K)$ during iteration b , and 0 otherwise. The perturbation $\epsilon_{i,b}$ is the Gaussian white noise added to the log-fold change of protein i , scaled by a hyperparameter η and the protein’s empirical standard error SE_i^2 . This ensures that noise injection is heteroskedastic and respects the instrument’s measurement uncertainty. The pathway’s overall stability, $\bar{S}_{lesi}(K)$, is calculated as the mean S_{lesi} of its core proteins.

Step 4: Topological Specificity via Information Content

Biological ontologies contain terms of wildly varying granularity, meaning enrichment scores alone cannot distinguish between broad, uninformative functional categories and highly targeted enzymatic cascades. We quantify the topological specificity of each pathway based on its occurrence probability within the global network, mathematically penalizing redundant parent nodes.

$$IC(K) = -\log_2 \left(\frac{|K|}{N_{proteome}} \right)$$

$IC(K)$ is the Information Content of pathway K . The denominator $N_{proteome}$ represents the total number of annotated proteins measured in the experimental universe, while the numerator $|K|$ is the number of proteins specifically mapped to pathway K . This ratio defines the prior probability $P(K)$ of a random protein belonging to the pathway. Taking the negative base-2 logarithm transforms this probability into a measure of entropy or specificity, where smaller, highly focused sub-pathways yield large IC values.

Step 5: Semantic Similarity and Medoid Reduction

To compress the highly redundant structure of the Gene Ontology into an orthogonal basis set, pathway enrichment, structural stability, and topological specificity are combined into a single fitness metric. Using semantic distances between pathway pairs, a weighted clustering algorithm extracts the optimal medoids, reducing the topological space to its most robust, non-overlapping Eigen-Pathways.

$$W(K_i) = |NES(K_i)| \cdot IC(K_i) \cdot \bar{S}_{lesi}(K_i) \quad \text{and} \quad \mathbf{S}_{Lin}(K_i, K_j) = \frac{2 \cdot IC(\text{MICA}(K_i, K_j))}{IC(K_i) + IC(K_j)}$$

$W(K_i)$ is the composite fitness weight of pathway K_i , factoring in the absolute Normalized Enrichment Score ($|NES|$), Information Content (IC), and mean stability ($\bar{S}_{lesi}$). $\mathbf{S}_{Lin}(K_i, K_j)$ is the Lin semantic similarity matrix capturing the biological overlap between pathways K_i and K_j . The term $\text{MICA}(K_i, K_j)$ refers to the Most Specific Common Ancestor shared by the two pathways in the ontology directed acyclic graph. Clustering operates on the distance matrix $\mathbf{D} = 1 - \mathbf{S}_{Lin}$, weighted by $W(K_i)$, to isolate N independent exemplar medoids representing the Eigen-Pathways.

Step 6: Asymmetric Directed Pathway Overlap Network

Rather than linking pathways symmetrically, which produces uninterpretable network clusters, the relationships between the extracted Eigen-Pathways are mapped as a bipartite projection. By calculating the directional subset overlap, the pipeline automatically infers hierarchical biological flow, generating a Directed Acyclic Graph capable of identifying critical signaling bridges.

$$\mathbf{W}_{A \rightarrow B} = \frac{|A \cap B|}{|A|} = P(B \mid A)$$

$\mathbf{W}_{A \rightarrow B}$ is the asymmetric directed edge weight representing the transition probability from Eigen-Pathway A to Eigen-Pathway B . The numerator $|A \cap B|$ is the precise integer count of significant proteins shared between the two pathways. The denominator $|A|$ is the total number of significant proteins annotated to the source pathway A . This normalization ensures that if a specialized sub-complex is entirely contained within a broader metabolic pathway, the directed edge weight reaches exactly 1.0, defining a strict structural hierarchy and breaking the symmetry ($\mathbf{W}_{A \rightarrow B} \neq \mathbf{W}_{B \rightarrow A}$).